Thesis Coordination System

AUTHORS & INSTITUTION

Afiya Humaira, Md Shariear Hossien Dipu

Department of CSE, Green University of Bangladesh

ABSTRACT

This project presents a Web-Based Thesis Management System designed to simplify and organize the academic thesis process. The system allows students to form teams, select supervisors, and submit required documents online. Supervisors and board members can review submissions, provide feedback, and manage evaluations efficiently. The platform improves communication, reduces manual work, and ensures transparency throughout the thesis lifecycle. It offers a secure and structured digital solution for academic project management.

INTRODUCTION

- The system provides a centralized web platform connecting students, supervisors, and board members.
- It simplifies team formation, supervisor selection, and document submission processes
- It ensures transparent communication, efficient workflow, and reduced manual effort in thesis management.

PROBLEM STATEMENT

- Manual handling of thesis activities leads to delays, errors, and poor coordination.
- Lack of a centralized system makes communication between students, supervisors, and boards inefficient.
- Difficulty in tracking thesis progress, submissions, and approvals causes confusion.
- Absence of proper digital records increases the risk of data loss and mismanagement.

MOTIVATION

- To simplify and automate the thesis management process for students and faculty.
- To reduce manual workload and improve coordination among all stakeholders.
- To create a transparent, efficient, and reliable academic workflow system.

OBJECTIVE

- To create a centralized web-based system for managing all thesis activities efficiently.
- To streamline team formation, supervisor allocation, document submission, and communication among users.
- To ensure transparency, reduce manual effort, and improve tracking of thesis progress and evaluations.

BACKGROUND STUDY

- Existing systems lack full integration for team formation, supervisor assignment, and defense scheduling (Kumar & Sharma, 2020).
- Centralized web platforms are needed to track thesis progress and submissions efficiently (Ali et al., 2019).
- Role-based access improves transparency and reduces administrative workload (Chen & Lee, 2021).
- Digital platforms enhance accountability, workflow, and provide real-time notifications (Rahman et al., 2022).

SDLC MODEL SELECTION

Table 1: Software development life cycle Table

Priority	Criteria	Waterfall	V-Shape	Iterative	Spiral	Agile	Prototype
5	Requirement changes	No	No	Yes	Yes	Yes	Yes
5	High team collaboration	No	No	Yes	No	Yes	No
4	Fast deployment and feedback loop	No	No	Yes	No	Yes	Yes
4	User-centric features like messaging	No	No	No	No	Yes	Yes
4	Early prototype/demo requirement	Yes	Yes	Yes	Yes	Yes	Yes
3	Risk management	No	No	Yes	Yes	Yes	No
3	Lightweight documentation	No	No	Yes	Yes	Yes	Yes
5	Supervisor selection and team forming	No	No	No	No	Yes	No
4	Time-bound delivery and sprints	No	No	No	No	Yes	No
	Total (Out of 37)	**4**	**4**	**24**	**15**	**37**	**20**

Selected model: Agile

HIGH LEVEL ARCHITECTURE

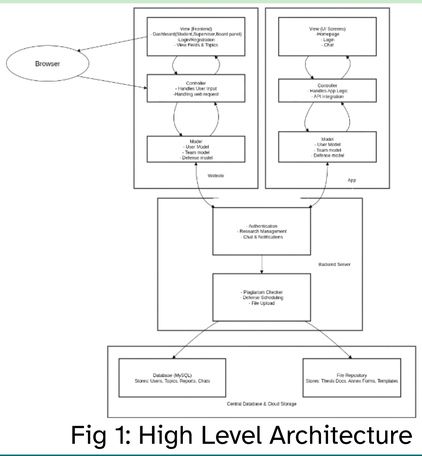

Fig 1: High Level Architecture

LOW LEVEL ARCHITECTURE

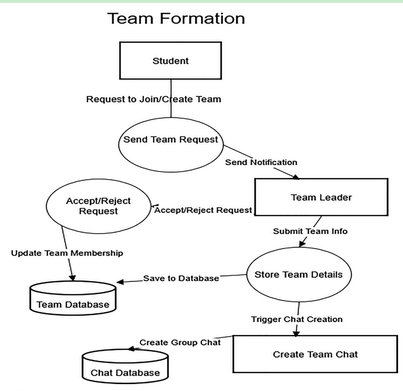

Fig 2: Low Level Architecture

USER INTERFACE DESIGN

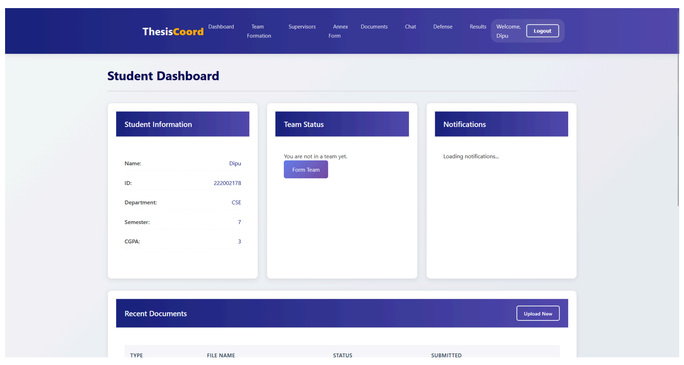

Fig 3: User Interface

TOOLS AND TECHNOLOGY

- Backend: Node.js, Express.js
- Database: MongoDB with Mongoose ODM
- Frontend: EJS (Templating), Custom CSS with Professional Design
- Authentication & Security: Session-based authentication with express-session, bcrypt.js for password hashing

TESTING

Table 2: Manual Testing Table

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result
Public Access	1) Unregistered user views Research Fields on Homepage.	URL: /index.html	Research fields and supervisor counts are displayed.	Research fields and supervisor counts are displayed.	Pass
Registration	2) Student registers with a unique ID and Email.	ID: 222002120 Email: mddipu1339@gmail.com	Registration successful; Redirect to Login.	Registration successful; Redirect to Login.	Pass
Registration	3) User attempts to register with an existing Email.	Email: afiya@gmail.com	Error: "User already exists".	Error: "User already exists".	Pass
Login	4) Student logs in with valid credentials.	Role: Student Pass: 123456	Redirect to Student Dashboard.	Redirect to Student Dashboard.	Pass
Login	5) Supervisor logs in with invalid password.	Role: Supervisor Pass: wrongpass	Error: "Invalid credentials".	Error: "Invalid credentials".	Pass

IMPLEMENTATION

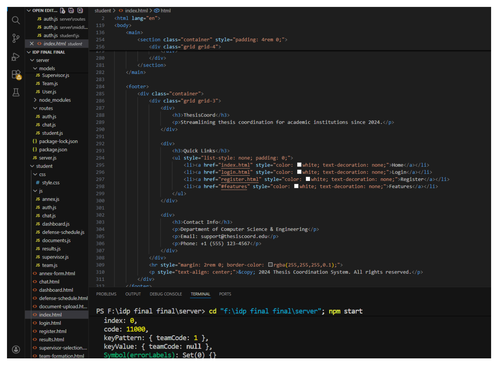

Fig 4: Implementation

SOCIAL IMPACT

- To simplify and automate the thesis management process for students and faculty.
- To reduce manual workload and improve coordination among all stakeholders.
- To create a transparent, efficient, and reliable academic workflow system.

CONCLUSION

- The system centralizes thesis management and minimizes manual processes.
- Role-based access ensures secure, structured, and controlled operations.
- Automated workflows improve tracking of teams, supervisors, and submissions.

REFERANCES

- Kumar, R., & Sharma, V. (2020). Web-based Academic Management Systems: Challenges and Opportunities. International Journal of Computer Applications, 176(12), 30–37.
- Ali, S., Ahmed, M., & Khan, T. (2019). Centralized Platforms for Thesis Monitoring and Management. Journal of Educational Technology, 15(2), 10–18.
- Chen, L., & Lee, K. (2021). Role-based Access Control in Academic Web Systems. International Journal of Information Management, 55, 102–110.
- Rahman, F., Hossain, S., & Islam, R. (2022). Digital Thesis Coordination Systems for Universities: A Case Study. Education and Information Technologies, 27(5), 5679–5695.